Exam Design *Intake*

A structured specification for the multi-agent exam generator.

INSTRUCTOR: DR. M. OKAFOR · COURSE: CHEM 3320 · TERM: FALL 2026 · SUBMITTED: 19 MAY 2026

§ 01 Course Materials

UPLOADED FILES

PDF	Clayden, Greeves & Warren — Organic Chemistry, Ch. 1–8 (lecture excerpts)	14.2 MB
PDF	Lecture notes, Weeks 1–7 (annotated slides)	8.1 MB
DOCX	Problem sets 1–4 with worked solutions	2.4 MB
PDF	Lab manual — Experiments 1–3 (SN1/SN2 kinetics, recrystallization)	3.7 MB
PDF	Prior midterm — Fall 2024 (style reference only)	0.6 MB

§ 02 Course Specification

OUTCOMES & TOPICS

COURSE CODE

CHEM 3320

COURSE TITLE

Organic Chemistry I

COURSE LEARNING OUTCOMES (WITH TARGET BLOOM LEVEL)

CLO-1	Predict the products of substitution and elimination reactions, given substrate, nucleophile, and reaction conditions.	APPLY
CLO-2	Distinguish SN1, SN2, E1, and E2 mechanisms by analyzing kinetic and stereochemical evidence.	ANALYZE
CLO-3	Draw curved-arrow mechanisms for elementary organic reactions with correct electron flow and intermediates.	APPLY
CLO-4	Evaluate competing reaction pathways and justify the predominant product on thermodynamic or kinetic grounds.	EVALUATE
CLO-5	Interpret experimental kinetic data to infer mechanism and rate-determining step.	ANALYZE

TOPICS & RELATIVE WEIGHTS

Nucleophilic substitution (SN1, SN2)	0.30
Elimination reactions (E1, E2)	0.25
Stereochemistry & conformational analysis	0.20
Reaction kinetics & energy diagrams	0.15
Acid–base chemistry & leaving groups	0.10

GUIDING PRINCIPLES

Emphasize mechanistic reasoning over memorization. Items should reward students who can predict outcomes from first principles, not those who have memorized specific reactions. Avoid rote nomenclature beyond what is required to communicate mechanism. Include at least two items based on interpretation of experimental data. No "trick" stereochemistry — students should be tested on understanding, not on visual puzzles.

§ 03 Exam Specification

FORMAT & STRUCTURE

EXAM TYPE

TOTAL POINTS

TIME BUDGET (MIN)

Midterm	100	75

ITEM TYPE DISTRIBUTION

MULTIPLE CHOICE	SHORT ANSWER	PROBLEM	DERIVATION	DATA INTERP.
10	4	3	2	2

TARGET DIFFICULTY DISTRIBUTION

EASY · 30%	MEDIUM · 50%	HARD · 20%

ACCOMMODATIONS REQUIRED

- [x] Extended-time variant (1.5×)
- [x] Screen-reader variant
- [x] Large-print variant
- [ ] Reduced-distraction format
- [x] Alt-text for all figures
- [ ] Translated variant

EQUATION RENDERING

LaTeX (MathJax)

FIGURES & DATA TABLES

Required (structures, kinetic plots)

§ 04 Trade-off Policy

RANKED PRIORITIES

Consulted when agents disagree. Higher rank wins. Reorder by drag-and-drop.

1 **Cognitive alignment**
Items must target the cognitive level promised by the course outcome.

2 **Content fidelity**
Items must be grounded in the uploaded course materials.

3 **Accessibility**
Plain language where possible; no construct-irrelevant difficulty.

4 **Discrimination**
Items separate students who understand from those who do not.

5 **Brevity**
Concise stems; minimal extraneous text.

§ 05 Refinement Parameters

LOOP CONTROL

MAXIMUM EPOCHS

4

CONVERGENCE RULE

No critical/high objections for one full epoch

MINIMUM OBJECTIONS PER CRITIC (EPOCH 1)

1

ESCALATE TO INSTRUCTOR AFTER

2 rebut-reassert cycles

Once submitted, the system will proceed to Phase 1: Blueprinting. You will be notified when the blueprint is ready for review.